

Program : Diploma in Wood and Paper Technology	
Course Code : 3124	Course Title: Mechanical and Electrical Engineering
Semester : 3	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To introduce the students the basic principles of Mechanical Engineering and apply the knowledge of these principles to the field of wood, paper and allied industries
- To impart a basic knowledge in electrical engineering with conceptual understanding of fundamental laws, machines and electronic components

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics I&II	1 & 2
Basic Knowledge of Applied Physics		Applied Physics I &II	1 & 2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the basic principles of fluid mechanics, principles of operations of pumps and power transmission	18	Applying
CO2	Illustrate the various properties of steam and working of steam generators.	11	Applying
CO3	Discuss the principles of D.C Circuits and A C Fundamentals	15	Applying
CO4	Summarize Electrical Machines and Electrical measuring Instruments	14	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Discuss the basic principles of fluid mechanics and principles of operations of pumps& transmission of power		
M1.01	Discuss the properties fluid and the working principles of pressure measuring devices	6	Understanding
M1.02	Explain the equations of fluid flow and its applications	6	Applying
M1.03	Outline the working principles and performance of centrifugal pumps and reciprocating pump	3	Applying
M1.04	Illustrate power transmission	3	Understanding

Contents:

Fluids - Definition - Properties - Density - Specific weight - specific volume - specific gravity - Viscosity - Vapor pressure - compressibility - Surface tension - Capillarity - Newtonian and Non-Newtonian fluids - pressure - Atmospheric pressure - Absolute pressure - Gauge pressure and vacuum pressure- Pressure measurement - piezometer, simple U-tube manometer, differential U- tube manometer; Buoyancy

Types of flow - Laminar flow - Turbulent flow - Steady flow - Unsteady flow - Uniform flow - Non Uniform flow - mean velocity - Continuity equation - Bernoulli's equation - (No Proof) - Applications - Venturimeter - Orifice meter - Pitot tube -Problems.

Flow through pipes - frictional losses in pipes - Darcy's formula (no derivation) and Chezy's equation.

Fans, blowers and compressors , pumps- reciprocating and centrifugal pumps

Transmission of power-beltdrives, open closed, length of belt, slip, Applications, problems

Gears; spur gear nomenclature (sketch required), velocity ratio -Types &applications

CO2	Illustrate the various properties of steam and working of steam generators.		
M2.01	Describe the steam properties and calculate the heat required for steam generation	4	Applying
M2.02	Explain boiler accessories and economizers	4	Understanding
M2.03	Illustrate the power generation in industries	3	Understanding
	Series Test - I	1	
Contents: Properties of steam - Steam formation under constant temperature - T-S curve, H-S Curve. Wet steam, dry steam - dryness fraction - super heated steam. Properties of steam - Latent heat, Specific heat, C_p , C_v , Enthalpy, Entropy, Specific volume. Application of steam tables - calculation of heat required for producing steam at a given pressure and temperature (Basic problems) Steam Generators and boilers: Introduction - Classification - fire tube and water tube boilers - Cochran boiler, Bob cock and Wilcox boiler - modern high-pressure boiler - La - Mont boiler. Boiler Mounting and accessories Thermal power generation with sketch			
CO3	Discuss the principles of D.C Circuits and A C Fundamentals		
M3.01	Explain the fundamental concepts of DC and AC electric circuits	8	Applying
M3.02	Summarize basic power system structure and laws	7	Applying
Contents: Basic principles of dc circuits - current, voltage, Power - definition and units. Resistance - laws of resistance - Ohm's Law - resistance in series and parallel - simple problems - Faraday's law of electromagnetic induction concept of self and mutual inductance units work and energy with simple problems AC Fundamentals - Production of sinusoidal voltage, definition of cycle, frequency, time period RMS value, average, form factor etc. Comparison of three phase and single phase AC Capacitor - capacitance in series and parallel. Concept of impedance in R.L.C series circuits. Power and power factor. AC circuits - simple problems.			
CO4	Summarize electrical machines and fundamentals of electronics		
M4.01	Discuss the construction, working principle and applications of simple DC Generator, DC Motor and Transformer	3	Understanding

M4.02	Explain the concept of three phase AC system	3	Understanding
M4.03	Summarize the construction, working principle and applications of various AC Machines used in wood based industries such as three phase induction Motor and single phase induction Motors.	5	Understanding
M4.04	Outline the fundamentals of electronic devices	3	Understanding
	Series Test - II	1	

Contents:

Electrical Machines

Transformers: - construction details - classification of transformers according to core fabrication EMF equation - losses, efficiency. Working principles of auto transformer - fields of application.

DC Motors - constructional details working principle, Types -series, shunt component and fields of application of each type.

Three phase induction motors - working principles. Types and fields and application

Single phase induction motors - types and method of starting, fields of application

Dynamometer type watt meter (sketches and brief explanation only)

Measuring instruments- moving coil and moving iron ammeters and voltmeters.

Electronics: - electron emission - P - type and N - type material, PN junction - characteristics of diode. Forward and reverse biasing - applications, Logic Gates: Symbol, Truth table and Boolean Expression of AND, OR, NOT, NAND, NOR gates. Microprocessor: Definition of microprocessor- Block diagram with pin details - uses.

Text / Reference:

T/R	Book Title/Author
T1	Abdulla Sheriff, Theory of Machines, Dhanapat Publications
R1	Khurmi, Theory of machines, S Chand
R2	Fluid Mechanics and Hydraulic machines by R K Banzal\
R3	B.L Theraja, Electrical Technology (Vol. -I), S Chand
R4	B.L Theraja, Electrical technology (Vol. - II), S Chand
R5	R.S.Khurmi, Thermal Engineering, S Chand
R6	B.L. Theraja, Electrical technology Vol. I & II, S Chand

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/112108148/
2	https://nptel.ac.in/courses/112105183/
3	https://nptel.ac.in/courses/108106072/
4	https://nptel.ac.in/courses/108106071/
5	https://nptel.ac.in/courses/108105017/